

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi-590018, Karnataka



A Mini Application Report on “Emotion Detection System”

Submitted in partial fulfilment of the requirements for **Machine Learning and Data**

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of

MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that the work entitled “**Emotion Detection System**” has been successfully carried out by **Mr. Rohith Gowda S (4AD25MC064)** as a part of the Skill Development Activities (SDA) component of the **Machine Learning and Data Analytics using Python (MMC201)** course during the academic year 2025–2026. The work submitted is a bona fide record of the student's effort and has been completed under the guidance and supervision of the faculty concerned. The student has satisfactorily demonstrated the implementation of concepts and techniques related to the course requirements. This report has been evaluated and approved as fulfilling the academic requirements prescribed for the Skill Development Activities (SDA) component of the Java Programming course.

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Emotion Detection System using Machine Learning

1. Introduction

Emotion Detection is a branch of Artificial Intelligence that enables a system to identify human emotions such as happiness, sadness, anger, fear, surprise, and neutral state from input data. These inputs can be facial expressions, voice signals, or textual data.

With advancements in Machine Learning (ML) and Deep Learning (DL), emotion recognition has become widely used in applications such as human-computer interaction, mental health monitoring, customer feedback analysis, virtual assistants, and security systems.

The objective of this project is to develop a real-time Emotion Detection System that identifies human emotions from facial expressions (or voice, depending on implementation) using Machine Learning techniques.

The system captures input (image or audio), preprocesses it, extracts meaningful features, and classifies the emotion using trained ML/DL models. The application is deployed using Python and Flask, providing a simple web interface for real-time emotion prediction. This project demonstrates the complete pipeline of an AI system including dataset handling, preprocessing, feature extraction, model training, evaluation, and deployment.

2. Problem Statement

Human emotions play a crucial role in communication, but traditional computer systems cannot understand emotional states. Most existing systems rely on manual feedback or are dependent on cloud-based APIs.

This project aims to design and implement a lightweight, real-time Emotion Detection System that can automatically recognize human emotions using Machine Learning. The system should capture input data (face image or voice), preprocess it, extract relevant features, classify emotions, and display results through a Flask web application. The goal is to create an efficient, accurate, and easily deployable system that works in real time.

3. System Requirements

Software Requirements

- Operating System: Windows 10/11
- Programming Language: Python 3.11+
- Framework: Flask
- Machine Learning Library: Scikit-learn / TensorFlow / Keras
- Image Processing: OpenCV
- Numerical Computing: NumPy
- Data Handling: Pandas
- Model Storage: Joblib / Pickle
- Frontend: HTML5, CSS3, JavaScript
- IDE: Visual Studio Code
- Browser: Chrome / Edge

Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: 5 GB free space
- Webcam (for facial emotion detection)
- Microphone (optional for voice-based system)
- Internet connection (optional)

4. Methodology

Dataset Description

The dataset consists of facial images (or audio samples) labeled with emotions such as:

- Happy
- Sad
- Angry
- Fear
- Surprise
- Neutral
- Disgust

For facial emotion detection, datasets like FER-2013 or self-collected images are commonly used. Each image is resized and converted into grayscale for uniform processing.

Algorithm Used

The Emotion Detection System uses supervised Machine Learning and Deep Learning techniques to classify emotions from input data.

The system processes images/voice, extracts features, and classifies emotions using models such as:

- Convolutional Neural Network (CNN)
- Support Vector Machine (SVM) (optional baseline)
- Random Forest (optional baseline)
- Multi-Layer Perceptron (MLP)

Step 1: Data Collection

- Images captured using a webcam or dataset files
- Each image labeled with corresponding emotion
- Audio samples (if voice-based system)

Step 2: Data Preprocessing

For image-based system:

- Resize images (e.g., 48×48 or 100×100)
- Convert to grayscale
- Normalize pixel values (0–1 scaling)
- Face detection using Haar Cascade or DNN

For audio-based system (optional):

- Noise removal
- Silence trimming
- Normalization

Step 3: Feature Extraction

For image-based system:

- Pixel intensity values
- Facial landmarks (optional)
- Deep features using CNN layers

For audio-based system:

- MFCC features
- Chroma features
- Spectral features

Step 4: Feature Scaling

- Normalize feature values for better convergence
- StandardScaler or MinMaxScaler is applied

Step 5: Model Training

1. Convolutional Neural Network (CNN) using TensorFlow/Keras

The Emotion Detection System uses a **Convolutional Neural Network (CNN)** built using **TensorFlow/Keras** for facial emotion recognition. CNN is highly effective for image-based classification tasks because it can automatically learn spatial features such as edges, textures, and facial patterns directly from input images.

The model is trained on preprocessed facial images labelled with different emotions such as Happy, Sad, Angry, Fear, Surprise, Neutral, and Disgust.

Step 6: Model Selection / Ensemble

The system selects the best-performing model (usually CNN). Optionally, ensemble learning can be used to combine predictions.

Step 7: Emotion Prediction

Workflow:

1. Capture image from webcam
2. Detect face region
3. Preprocess image
4. Extract features
5. Predict emotion using a trained model
6. Display result on Flask interface

Step 8: Output Classes

The system predicts one of the following emotions:

- Happy
- Sad
- Angry
- Fear
- Surprise
- Neutral

Confidence Score

Each prediction includes probability values for all emotions. The emotion with the highest probability is selected.

Example:

Emotion Probability

Happy 82.45%

Sad 70.32%

Angry 73.11%

Neutral 78.12%

If confidence is below threshold, output is marked as **Unknown**.

Advantages of Proposed System

- Real-time emotion recognition
- Works with webcam input
- High accuracy using CNN
- Easy Flask integration
- Can be extended to multi-modal (face + voice)
- Scalable for real-world applications

5. System Architecture

The system architecture includes:

- Input Layer: Webcam / Audio input
- Preprocessing Module: Face detection / audio cleaning
- Feature Extraction: CNN layers / MFCC
- Classification Model: Trained ML/DL model
- Flask Backend: Handles prediction requests
- Frontend: Displays emotion result

7. Code Snippet

Model Train

CNN Model

```
model = Sequential([  
  
    Conv2D(32, (3,3), activation='relu', input_shape=(48,48,1)),  
  
    BatchNormalization(),  
  
    MaxPooling2D(2,2),  
  
  
    Conv2D(64, (3,3), activation='relu'),  
  
    BatchNormalization(),  
  
    MaxPooling2D(2,2),  
  
  
    Conv2D(128, (3,3), activation='relu'),  
  
    BatchNormalization(),  
  
    MaxPooling2D(2,2),  
  
    Flatten(),  
  
    Dense(256, activation='relu'),  
  
    Dropout(0.5),  
  
    Dense(5, activation='softmax')  
  
])  
  
model.compile(
```

```
optimizer='adam',

loss='categorical_crossentropy',

metrics=['accuracy']

)

model.summary()

model.fit(

    train_data,

    validation_data=test_data,

    epochs=EPOCHS

)

os.makedirs('model', exist_ok=True)

model.save('model/emotion_cnn.keras')

print("✔ Model saved to model/emotion_cnn.keras")
```

Flask Route

```
@app.route("/health")

def health():

    return {

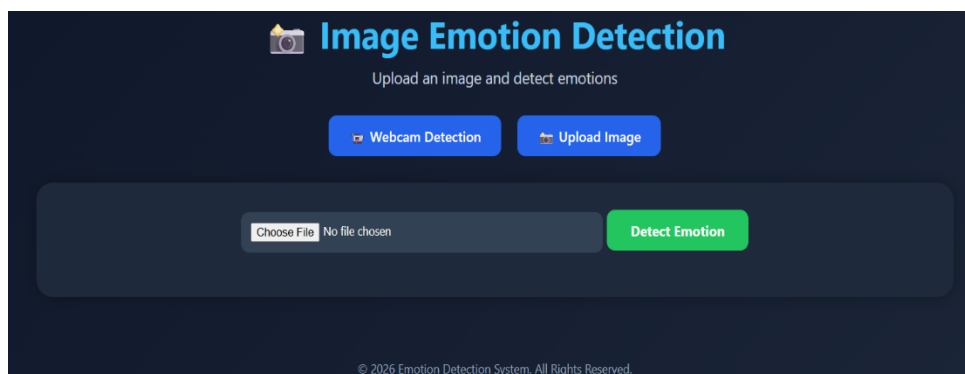
        "status": "running",

        "camera": camera.isOpened()

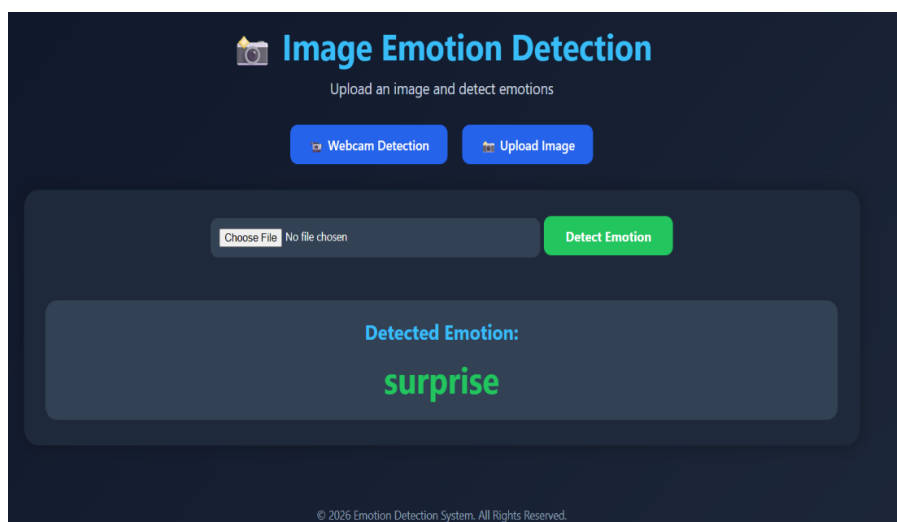
    }
```

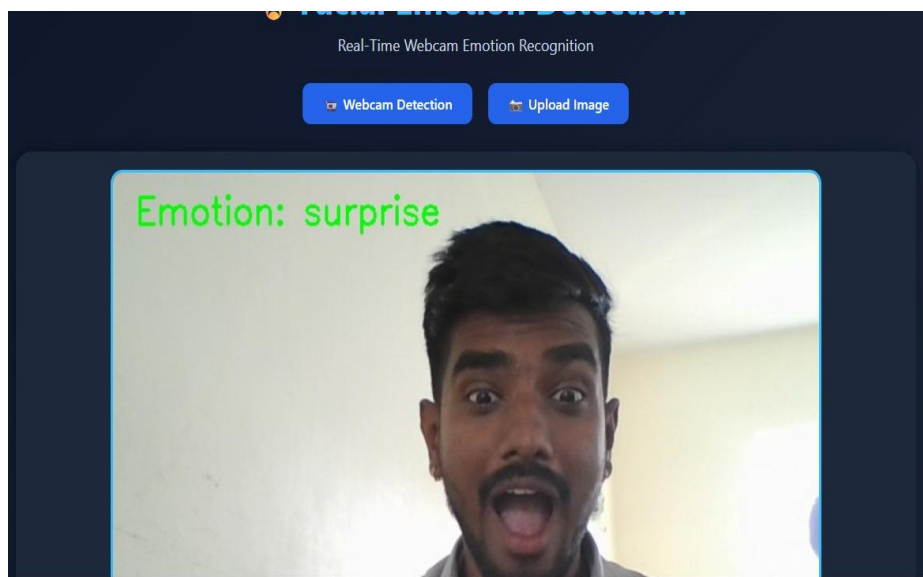
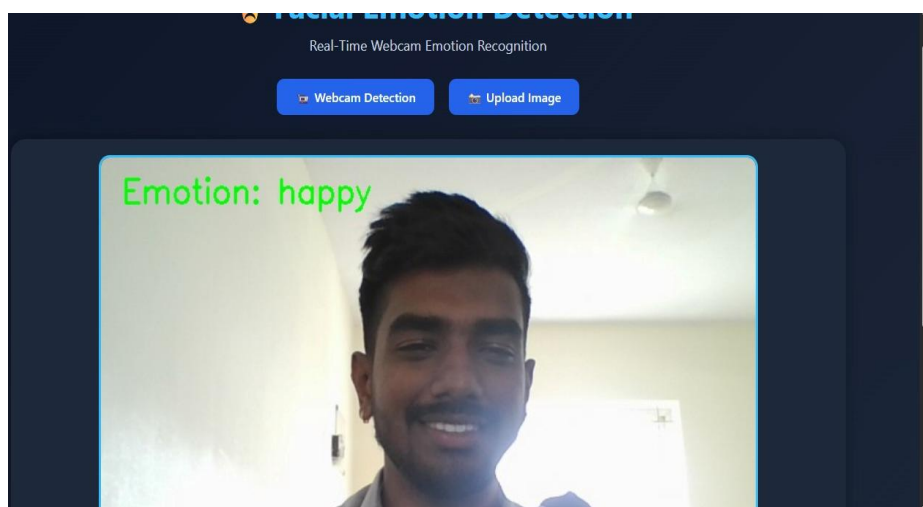
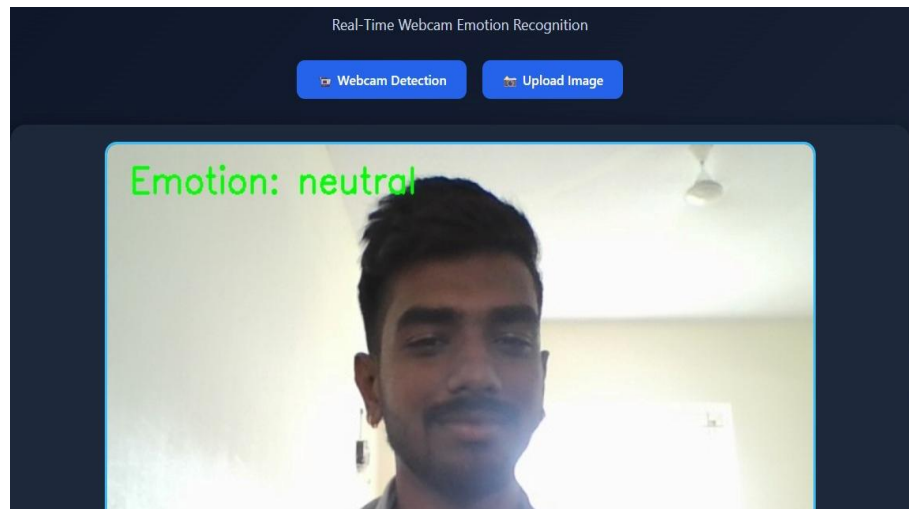
```
if __name__ == "__main__":  
  
    Timer(2, open_browser).start()  
  
    app.run(  
  
        host="0.0.0.0",  
  
        port=5000,  
  
        debug=False  
    )
```

8. Snapshot



After detection:





8. Conclusion

The Emotion Detection System using Machine Learning successfully demonstrates how human emotions can be identified using artificial intelligence techniques. The system integrates image preprocessing, feature extraction, deep learning classification, and Flask deployment into a unified application.

The model effectively detects facial emotions in real time and provides accurate predictions with confidence scores. The system is useful for human-computer interaction, sentiment analysis, and behavioral studies.

Future Scope

- Improve accuracy using advanced CNN architectures (ResNet, VGG)
- Multi-modal emotion detection (face + voice + text)
- Real-time mobile application deployment
- Integration with virtual assistants
- Support for more emotion categories
- Better performance in low-light and noisy environments